**EDA Summary Report**

**Project Title**

**Wine Quality Exploratory Data Analysis**

**Objective**

The main objective of this project was to understand the Wine Quality dataset by performing Exploratory Data Analysis (EDA). I explored the dataset, cleaned the data, calculated statistical measures, analyzed the relationship between different features, detected outliers, and created visualizations to better understand the data.

**Dataset Description**

The Wine Quality dataset contains different chemical properties of wine such as fixed acidity, volatile acidity, citric acid, alcohol content, pH, sulphates, density, and quality rating. These features help in analyzing the factors that influence the quality of wine.

**Data Exploration**

I first explored the dataset using head(), info(), shape, and describe() to understand the structure of the data. This helped me identify the number of rows, columns, data types, and summary statistics.

**Data Cleaning**

I checked the dataset for missing values using isnull().sum(). Some numerical columns contained missing values, so I replaced them with their **median** values because the median is less affected by outliers. I also checked for duplicate records and removed them to improve data quality.

**Descriptive Statistics**

I calculated the mean, median, mode, variance, and standard deviation for the numerical columns. These statistical measures helped me understand the central tendency and spread of the data.

**Correlation Analysis**

I calculated the correlation matrix to understand how different variables are related to each other. From the analysis, I observed that:

- Alcohol has a positive correlation with wine quality (0.469555), which means wines with higher alcohol content generally have better quality ratings.
- Density has a strong negative correlation with alcohol (-0.668216), indicating that density decreases as alcohol content increases.
- Free sulfur dioxide and total sulfur dioxide have a strong positive correlation (0.720666), showing that these two variables increase together.
- Volatile acidity has a negative correlation with quality (-0.264280), suggesting that higher volatile acidity is associated with lower wine quality.

**Outlier Detection**

I detected outliers using the Interquartile Range (IQR) method. After calculating the lower and upper limits, I identified outliers in the selected numerical column and treated them using the capping method to reduce their effect on the analysis.

**Data Analysis**

I performed filtering and sorting operations to analyze wines based on quality and alcohol content. This helped me understand how different features vary across the dataset.

**Visualizations**

I created the following visualizations:

- Bar Chart
- Line Chart
- Histogram
- Scatter Plot

These graphs helped me understand the distribution of data and the relationship between different variables more clearly.

**Conclusion**

Through this project, I gained practical experience in Exploratory Data Analysis using Python. I learned how to clean a dataset, handle missing values, perform statistical analysis, identify correlations, detect outliers, and visualize data. Based on my analysis, alcohol content showed a positive relationship with wine quality, while volatile acidity showed a negative relationship. Overall, this project improved my understanding of data preprocessing and data analysis techniques.